Engineering Chemistry – Unit 1 Master

Engineering Chemistry (UE26CY151A) · Semester 1 · Unit 1 · Compiled from PESU AV-summary transcripts, class slides, the Unit-1 slide PDF and past ESA papers · 13 Oct 2026

★★★ high · ★★ medium · ★ low/conceptual weight | Not on current slides Transcript only Added for clarity | source tags: [S004·8] slide, [T005] transcript, [PDF] 137-page deck, [ESA] past paper

Contents

M1 · Molecular Spectroscopy
- T1 · EM radiation & interaction with matter
- T2 · Electromagnetic spectrum & regions
- T3 · Born–Oppenheimer approximation
- T3b · Beer–Lambert law
- T4 · Rotational (microwave) spectroscopy
- T5 · Vibrational (IR) – SHO
- T6 · Vibrational – anharmonic
- T7 · Electronic spectroscopy & Franck–Condon

M2 · Phase Equilibria
- T8 · Basic terms & phase rule
- T9 · Derivation of phase rule
- T10 · One-component system: water
- T11 · Two-component: reduced phase rule & Pb–Ag
- T12 · Thermal analysis / cooling curves

M3 · Computational Chemistry
- T13 · Computational chemistry

Appendix A

Appendix B

Appendix C

Appendix D

M1 · Molecular Spectroscopy

T1 · EM radiation & interaction with matter

Short note

Spectroscopy studies radiation–matter interaction. A spectrophotometer has source, monochromator, sample, detector, recorder. EM waves have E ⊥ B ⊥ direction. $E = h\nu$, $\nu = c/\lambda$, $\bar{\nu} = 1/\lambda$. Atomic spectra are lines; molecular spectra are bands.

Formulas

F1 · Photon energy ★★★

$$E = h\nu$$

Gives Energy of radiation · **Applies** any EM radiation · **Units** J · [S001·5]

F2 · Frequency–wavelength ★★★

$$\nu = c/\lambda$$

Gives Frequency · **Applies** vacuum/air · **Units** Hz · [S001·5]

F3 · Wavenumber ★★★

$$\bar{\nu} = 1/\lambda \; ; \; E = hc\bar{\nu}$$

Gives Wavenumber; energy · **Applies** spectroscopic unit · **Units** cm^{-1} · [S001·5]

F4 · Transition energy ★★★

$$\Delta E = E_2 - E_1 = h\nu$$

Gives Absorbed/emitted quantum · **Applies** two quantised levels · **Units** J (÷hc → cm^{-1}) · [S001]

Important points

Definition: Molecular spectroscopy: the branch of science dealing with the interaction of electromagnetic radiation with matter (molecules). [S001·3] ★★

Spectrophotometer: source → monochromator (entrance slit, dispersive element, exit slit) → sample → detector → recorder. [S001·3] ★★

EM radiation has dual character (photons moving as waves); E and B fields oscillate perpendicular to each other and to the direction of propagation; all travel at the speed of light. [S001·4] ★★

Atomic spectra: electronic transitions between atomic orbitals → line spectra. Molecular spectra: electronic (HOMO→LUMO) plus vibrational and rotational transitions → band spectra. [S001·6-7] ★★

Absorption: molecule goes to a higher level taking ΔE = hν; emission: falls to a lower level releasing hν. [S001] ★

Flowcharts & tables

Atomic vs molecular spectra

	Atomic	Molecular

	Atomic	Molecular
Transition	between atomic orbitals	HOMO→LUMO + vib + rot
Appearance	line spectra	band spectra

T2 · Electromagnetic spectrum & regions

Short note

Radio → NMR/ESR; microwave → rotation; IR → vibration; UV-Vis → valence electrons; X-ray → inner electrons; γ → nuclei. Raman: Rayleigh (no shift), Stokes (lower ν), anti-Stokes (higher ν).

Important points

Radio (10 m–1 cm): NMR, ESR – nuclear / electron spin reversal. Microwave (1 cm–100 μm): rotational. IR ★★★ (100 μm–1 μm): vibrational. UV-Vis (1 μm–10 nm): electronic, valence electrons. X-ray (10 nm–100 pm): inner electrons. γ (100–1 pm): nuclear rearrangement. [S002·4-5]

Why: Rayleigh scattering: scattered light has the same frequency as the incident light (elastic). Raman: shifted ★★ frequency — Stokes lines (lower frequency, molecule gains energy), anti-Stokes lines (higher frequency, molecule loses energy). [T002] Transcript only Added for clarity

Flowcharts & tables

Microwave vs IR vs UV-Vis

	Microwave	IR	UV-Vis
Transition	rotational	vibrational	electronic
Condition	permanent dipole	dipole changes	any molecule
Rule	$\Delta J = \pm 1$	$\Delta v = \pm 1$ (SHO)	none for v; Franck–Condon
Gives	B, I, r_o	k, $\bar{\nu}$	electronic states

Rayleigh vs Raman

	Rayleigh	Stokes	Anti-Stokes
Frequency	unchanged	lower	higher
Molecule	unchanged	gains energy	loses energy

Past-paper questions

(×2) Rayleigh vs Raman; Stokes vs anti-Stokes.

T3 · Born–Oppenheimer approximation

Short note

$E = E_{rot} + E_{vib} + E_{elec}$, $E_{rot} \ll E_{vib} \ll E_{elec}$, components independent; translation negligible.

Formulas

F5 · Total molecular energy ★★

$$E = E_{trans} + E_{rot} + E_{vib} + E_{elec} \approx E_{rot} + E_{vib} + E_{elec}$$

Gives Born–Oppenheimer · **Applies** E_trans negligible · **Units** J · [S003·4]

F6 · Ordering ★★

$$E_{rot} \ll E_{vib} \ll E_{elec}$$

Gives Microwave < IR < UV-Vis · **Applies** — · **Units** — · [S003·4]

F7 · Change in energy ★★

$$\Delta E = \Delta E_{elec} + \Delta E_{vib} + \Delta E_{rot}$$

Gives Electronic transition fine structure · **Applies** — · **Units** J or cm^{-1} · [S007·4]

Important points

Statement: *Born–Oppenheimer approximation: electronic, vibrational and rotational energies of a molecule are independent; E = E_rot + E_vib + E_elec (E_trans negligible).* [S003·4] ★★

$E_{rot} \ll E_{vib} \ll E_{elec}$ because electronic levels are farther apart than vibrational, which are farther than rotational. [S003·4] ★★

Exam trap: Mnemonic: Rot < Vib < Elec ↔ Microwave < IR < UV-Vis. [study aid] Added for clarity ★

Graphs

No graph for this topic (see formulas/notes).

Past-paper questions

(×3) State Born–Oppenheimer; energy-level diagram.

T3b · Beer–Lambert law

Short note

$A = \log(I_0/I) = \varepsilon cl$ (older syllabus; not on current slides).

Formulas

F8 · Beer–Lambert Not on current slides Added for clarity ★

$$A = \log(I_0/I) = \varepsilon cl$$

Gives Absorbance · **Applies** dilute solutions · **Units** dimensionless · [—]

Important points

Beer–Lambert law: absorbance $A = \log(I_0/I) = \varepsilon cl$ — proportional to concentration c and path length l; ε is the molar absorptivity. [ESA Dec-24] Not on current slides Added for clarity ★

Past-paper questions

Dec-24 (older syllabus) State the Beer–Lambert law.

T4 · Rotational (microwave) spectroscopy

Short note

Rigid rotor: two masses at fixed r_0. $I = \mu r_0^2$. $\varepsilon_J = BJ(J+1)$. Needs permanent dipole; $\Delta J = \pm 1$. Lines at $2B(J+1)$, spacing $2B$. From B get I and r_0.

Formulas

F9 · Centre of mass ★★★

$$m_1 r_1 = m_2 r_2 \; ; \; r_0 = r_1 + r_2$$

Gives Positions of atoms about C · **Applies** rigid diatomic · **Units** m · [S004; PDF]

F10 · Distances ★★★

$$r_1 = m_2 r_0/(m_1+m_2) \; ; \; r_2 = m_1 r_0/(m_1+m_2)$$

Gives — · **Applies** — · **Units** m · [PDF]

F11 · Moment of inertia ★★★

$$I = m_1 r_1^2 + m_2 r_2^2 = \mu r_0^2 \text{ with } \mu = m_1 m_2/(m_1+m_2)$$

Gives Moment of inertia · **Applies** rigid rotor · **Units** kg m² · [S004·7; PDF]

F12 · Rotational energy ★★

$$E_r = \tfrac{1}{2} I \omega^2 = L^2/2I$$

Gives Classical rotor · **Applies** — · **Units** J · [PDF]

F13 · Rotational levels (J) ★★★

$$E_J = h^2 J(J+1)/(8\pi^2 I)$$

Gives Quantised levels · **Applies** J = 0,1,2... · **Units** J · [S004·8]

F14 · Rotational levels (cm^{-1}) ★★★

$$\varepsilon_J = BJ(J+1), \; B = h/(8\pi^2 Ic)$$

Gives Level energy; rotational constant · **Applies** c in cm s^{-1} · **Units** cm^{-1} · [S004·8]

F15 · Selection rules ★★★

$$\Delta J = \pm 1 \; ; \text{ permanent dipole needed}$$

Gives Allowed transitions · **Applies** gross rule · **Units** — · [S004·10]

F16 · Line positions ★★★

$$\Delta\varepsilon(J \rightarrow J+1) = 2B(J+1) \; ; \text{ spacing} = 2B \; ; \text{ levels } 0,2B,6B,12B,20B,30B,42B$$

Gives Rotational spectrum · **Applies** — · **Units** cm^{-1} · [S004·9-10]

Important points

Definition: Gross selection rule (rotational): molecule must have a permanent dipole moment. Specific rule: $\Delta J = \pm 1$. [S004·3,10] ★★★

Why: HCl and CO are microwave-active (permanent dipole); H_2, Cl_2 and CO_2 are not (no permanent dipole). ★★★ [S004·5]

Rigid-rotor model: heteronuclear diatomic = two masses joined by a rigid massless rod; J = 0,1,2…; levels 0, 2B, 6B, 12B… [S004·6-9] ★★★

First line is at 2B (not B) and consecutive lines are 2B apart; B → I → r_0 via $I = \mu r_0^2$. [S004·12] ★★★

Intensity rises then falls with J because higher levels are populated at room temperature (thermal population). ★★ [S004·12]

Exam trap: Traps: J starts at 0; first line = 2B; use μ in kg (not amu); use c in cm s^{-1} when B is in cm^{-1}. ★★★ [exam]

Graphs

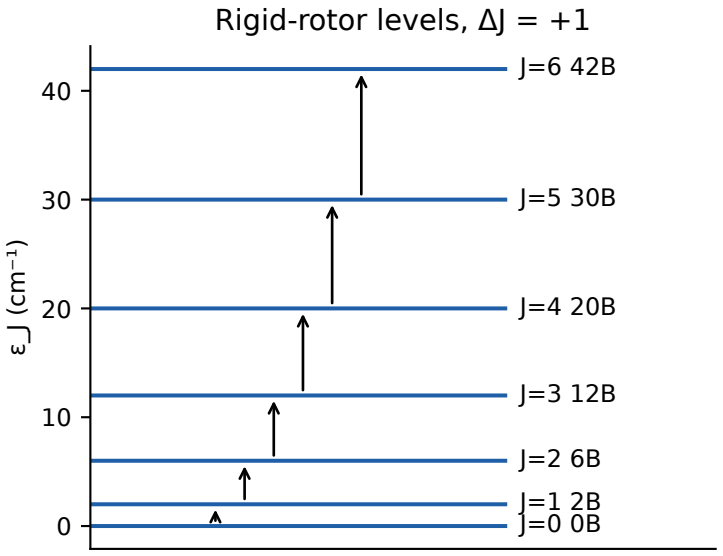


Fig. G9 — Rigid-rotor energy levels J=0–6 (values J(J+1)B), ΔJ=+1 arrows.

In an exam: Label J, the energy in B, and the widening gaps 2B,4B,6B…

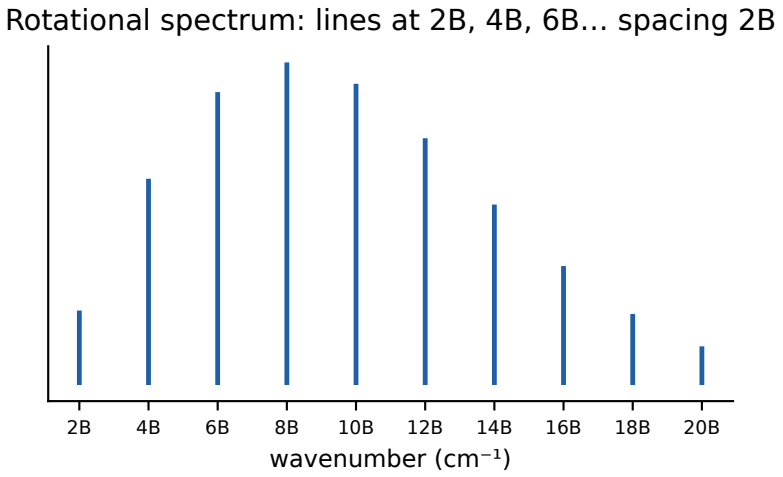


Fig. G10 — Rotational stick spectrum.

In an exam: Lines at 2B, 4B, 6B…; constant spacing 2B; intensity rises then falls.

Flowcharts & tables

F3 · Spectroscopic activity checker

Permanent dipole? → Yes: microwave (rotational) active

↓

Dipole changes during a vibration? → Yes: IR active

↓

All molecules → electronic (UV-Vis) active

↓

HCl: all three · CO: all three · H_2, Cl_2: electronic only · CO_2: IR (asym. stretch, bend) + electronic, not microwave · H_2O: all three · HI: all three

F4 · Rotational spectrum → bond length

Read first line (or spacing) → B = line/2

↓

$I = h / (8\pi^2 cB)$

↓

$\mu = m_1 m_2/(m_1+m_2)$ in kg

↓

$r_0 = \sqrt{(I/\mu)}$

Worked examples

Given	Solution	Answer
HCl: first rotational line at 21.18 cm^{-1} (H=1.008, Cl=35.45). Find B, I, r_0.	B = line/2 = 10.59 cm^{-1}; I = h/(8π²cB) = 2.64e-47 kg m²; μ = 1.63e-27 kg; $r_0 = \sqrt{(I/\mu)}$	B = 10.59 cm^{-1}, I = 2.64e-47 kg m², r_0 = 1.27 Å
HI: first line 13.022 cm^{-1} (H=1.0, I=126.9). Find I and r_0.	B = 6.511 cm^{-1}; I = h/(8π²cB); $r_0 = \sqrt{(I/\mu)}$	I = 4.30e-47 kg m², r_0 = 1.62 Å
B = 10.56 cm^{-1}. Find I and line spacing.	I = h/(8π²cB); spacing = 2B	I = 2.65e-47 kg m²; spacing = 21.12 cm^{-1}
HCl J=0→1 at 20.68 cm^{-1}. Find the J=20→21 line.	Δε = 2B(J+1) and 2B = 20.68	2B·21 = 21 × 20.68 = 434.28 cm^{-1}

Past-paper questions

2023–25 (×4) Derive $I = \mu r_0^2$ for a diatomic rigid rotor; calculate bond length from first line (HCl 21.18 cm^{-1}; HI 13.022 cm^{-1}); draw levels to J=5/6 and the spectrum.

T5 · Vibrational (IR) – SHO

Short note

SHO: spring/Hooke's law, parabolic PE. $\varepsilon_v = (v+½)\bar{\nu}$, ZPE = ½$\bar{\nu}$, equal spacing, $\Delta v=\pm1$ → one line at $\bar{\nu}$. Needs changing dipole; $\bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)}$.

Formulas

F17 · Hooke's law ★★

$$F = -kx \; ; \; PE = ½k(r - r_{eq})^2$$

Gives Parabolic potential · **Applies** SHO · **Units** N ; J · [S005·6-7]

F18 · Oscillation frequency ★★★

$$\nu_{osc} = (1/2\pi)\sqrt{(k/\mu)}$$

Gives Frequency · **Applies** SHO · **Units** Hz · [S005·8]

F19 · Wavenumber of vibration ★★★

$$\bar{\nu}_{osc} = (1/2\pi c)\sqrt{(k/\mu)}$$

Gives Line position · **Applies** c in cm s^{-1} · **Units** cm^{-1} · [S005·8]

F20 · SHO levels ★★★

$$E_v = (v+½)h\nu_{osc} \; ; \; \varepsilon_v = (v+½)\bar{\nu}_{osc}$$

Gives Levels · **Applies** v = 0,1,2… · **Units** J ; cm^{-1} · [S005·8-9]

F21 · Zero-point energy ★★★

$$\varepsilon_0 = ½\bar{\nu}_{osc}$$

Gives Lowest level (≠ 0) · **Applies** — · **Units** cm^{-1} · [S005·9]

F22 · Selection rule & spacing ★★★

$$\Delta v = \pm1 \; ; \; \text{spacing} = \bar{\nu}_{osc} \; ; \; \text{one line at } \bar{\nu}_{osc}$$

Gives — · **Applies** dipole must change · **Units** cm^{-1} · [S005·9]

F23 · Force constant ★★★

$$k = 4\pi^2c^2\bar{\nu}^2\mu$$

Gives Bond stiffness · **Applies** μ in kg · **Units** N m^{-1} · [derived]

Important points

Definition: Gross selection rule (IR): the vibration must change the dipole moment; $\Delta v = \pm1$. [S005·3,9] ★★★

Why: CO_2 has no permanent dipole (microwave-inactive) but is IR-active: asymmetric stretch and bending change the dipole; symmetric stretch does not. [S005·4-5] ★★★

Why: HI is IR-active but H_2 and I_2 are not (homonuclear: no dipole change). [S005·4] ★★

Why: SHO shows only one line: levels are equally spaced and $\Delta v = \pm1$, so every allowed transition has the same energy $\bar{\nu}_{osc}$. [S005·9] ★★★

Zero-point energy $\varepsilon_0 = ½\bar{\nu}_{osc} \neq 0$: the molecule never rests even at v = 0. Bond treated as a spring obeying Hooke's law → parabolic PE. [S005·6-9] ★★★

> Every bond has a characteristic frequency, so an IR spectrum is a molecular 'fingerprint'. [S005·6] ★

Graphs

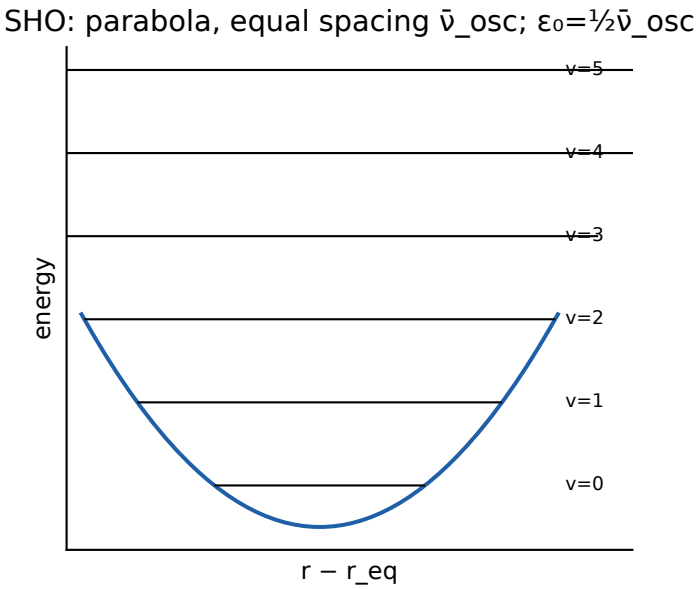


Fig. G13 — SHO parabola with equally spaced levels.

In an exam: Mark v=0…5 and $\varepsilon_0=\frac{1}{2}\bar{\nu}$; spectrum is a single line.

Flowcharts & tables

F5 · IR line → force constant (and reverse)

$\bar{\nu} \rightarrow k = 4\pi^2c^2\bar{\nu}^2\mu$

↓

$k \rightarrow \bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)} \rightarrow ZPE = \frac{1}{2}\bar{\nu} \rightarrow \varepsilon_v = (v+\frac{1}{2})\bar{\nu}$

Worked examples

Given	Solution	Answer
HF, k = 970 N m⁻¹ (H=1.008, F=18.99). Find $\bar{\nu}$ and zero-point energy.	$\bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)}$; ZPE = $\frac{1}{2}\bar{\nu}$ (× hc for joules)	$\bar{\nu}$ = 4148 cm⁻¹; ZPE = 2074 cm⁻¹ = 4.12e-20 J
CO, $\bar{\nu}$ = 2168 cm⁻¹ (SHO). Find k and ZPE.	$k = 4\pi^2c^2\bar{\nu}^2\mu$	k = 1.90e+03 N m⁻¹; ZPE = 1084 cm⁻¹
NO, k = 1530 N m⁻¹ (N=14, O=16). Find $\bar{\nu}$, ZPE, ε_3, ε_4.	$\varepsilon_v = (v+\frac{1}{2})\bar{\nu}$	$\bar{\nu}$ = 1865; ZPE = 933; ε_3 = 6528; ε_4 = 8393 cm⁻¹

Past-paper questions

2023–25 Calculate $\bar{\nu}$ and ZPE from k (HF, NO); k from $\bar{\nu}$ (CO); why only one line in SHO?

T6 · Vibrational – anharmonic

Short note

Morse potential: $\varepsilon_v = (v+½)\bar{\omega}_e - (v+½)^2\bar{\omega}_e x_e$. Levels converge; $\Delta v = \pm1, \pm2\ldots$; fundamental, overtones; hot bands left of fundamental.

Formulas

F24 · Morse function ★★

$$PE = D_{eq}[1 - \exp\{a(r_{eq} - r)\}]^2$$

Gives Anharmonic potential · **Applies** — · **Units** J · [S006·3]

F25 · Anharmonic levels ★★

$$\varepsilon_v = (v+½)\bar{\omega}_e - (v+½)^2\bar{\omega}_e x_e$$

Gives Converging levels · **Applies** x_e small, positive · **Units** cm^{-1} · [S006·4]

F26 · Transitions ★★

Fundamental $0 \to 1$: $\bar{\omega}_e(1-2x_e)$; 1st overtone $0 \to 2$: $2\bar{\omega}_e(1-3x_e)$; 2nd overtone $0 \to 3$: $3\bar{\omega}_e(1-4x_e)$

Gives — · **Applies** $\Delta v = \pm1, \pm2, \pm3\ldots$ · **Units** cm^{-1} · [S006·6-7]

F27 · Hot band & ZPE Added for clarity ★★

Hot band $1 \to 2$: $\bar{\omega}_e(1-4x_e)$ (left of fundamental) ; $\varepsilon_0 = ½\bar{\omega}_e(1-½x_e)$

Gives — · **Applies** higher T · **Units** cm^{-1} · [S006·9; ZPE added]

Important points

Why: Why SHO fails: real bonds dissociate when stretched; the true potential is anharmonic (Morse), so levels converge and overtones appear. [S006·3] ★★

Anharmonic: $\Delta v = \pm1, \pm2, \pm3\ldots$; most transitions start from v = 0 → fundamental ($0 \to 1$), first overtone ($0 \to 2$), second overtone ($0 \to 3$). [S006·6] ★★

Hot bands: from v = 1 at higher temperature; $1 \to 2$ appears to the left of the fundamental. [S006·9] ★★

x_e is small and positive; line spacing decreases with increasing v. [S006·4] ★★

Graphs

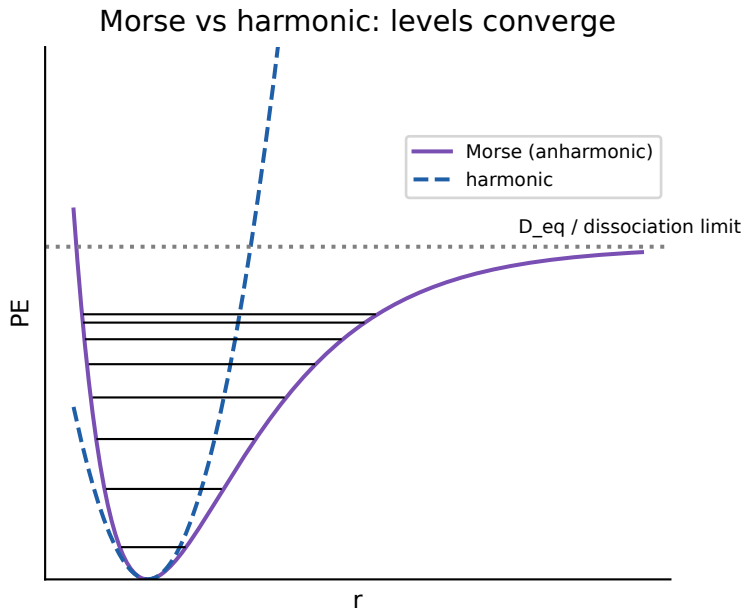


Fig. G15 — Morse curve vs harmonic parabola.

In an exam: Show D_eq, convergence of levels and dissociation limit.

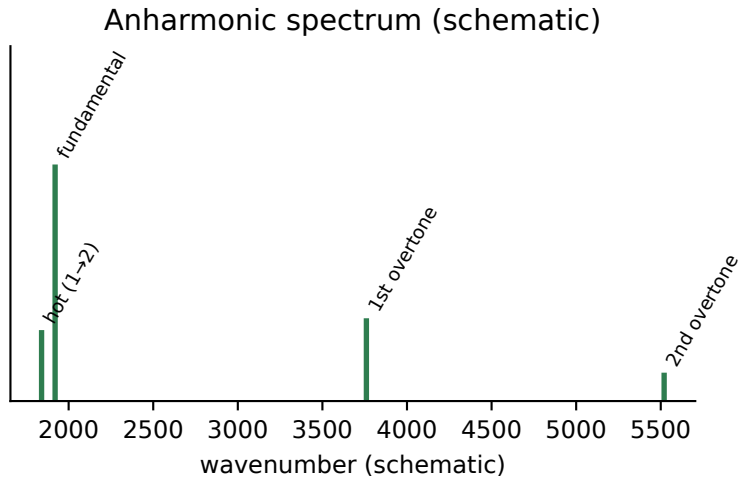


Fig. G17 — Anharmonic IR spectrum (schematic).

In an exam: Hot band left of fundamental; overtones with decreasing spacing.

Flowcharts & tables

F6 · Anharmonic solver

fundamental = $\bar{\omega_e}(1-2x_e)$; overtone = $2\bar{\omega_e}(1-3x_e)$

↓

ratio overtone/(2·fund) = $(1-3x_e)/(1-2x_e) \rightarrow x_e$

↓

$\bar{\omega_e}$ = fundamental/$(1-2x_e)$

SHO vs anharmonic

	SHO	Anharmonic

	SHO	Anharmonic
Potential	parabola	Morse
Levels	equal spacing	converge
Rule	Δv=±1	Δv=±1,±2,±3...
Spectrum	1 line	fundamental + overtones + hot bands

Worked examples

Given	Solution	Answer
HBr: fundamental 2559.08, first overtone 5027.54 cm^{-1}. Find x_e and $\bar{\omega}_e$.	fund = $\bar{\omega}_e(1-2x_e)$; overtone = $2\bar{\omega}_e(1-3x_e)$	x_e = 0.0171; $\bar{\omega}_e$ = 2649.7 cm^{-1}
CO: fundamental 2143 cm^{-1}, x_e = 0.01. Find $\bar{\omega}_e$.	$\bar{\omega}_e$ = fund/$(1-2x_e)$	$\bar{\omega}_e$ = 2186.7 cm^{-1}
CO: $\bar{\omega}_e$ = 2050, x_e = 0.017. First overtone?	$2\bar{\omega}_e(1-3x_e)$	3890.9 cm^{-1}

Past-paper questions

2023–25 Find x_e and $\bar{\omega}_e$ from fundamental and overtone; selection rule; failure of SHO.

T7 · Electronic spectroscopy & Franck–Condon

Short note

HOMO → LUMO transitions; vibrational coarse structure gives progressions (v′,v″); no vibrational selection rule; Franck–Condon: vertical transitions → intensity pattern by r_e' vs r_e''; continuum = dissociation.

Formulas

F28 · Electronic spectrum ★★

$$\varepsilon = \varepsilon_elec + \varepsilon_vib \text{ ; lines labelled } (v',v'')$$

Gives Progression · **Applies** no vibrational selection rule · **Units** cm^{-1} · [S007·6-8]

Important points

Electronic transitions: HOMO → LUMO (σ → σ*, π → π*, n → π*) in the UV-Vis; all molecules show them; visible transitions give colour. [S007·3] ★★★

Vibrational changes give 'coarse structure', rotational changes give 'fine structure'. Homonuclear molecules (H_2, N_2) show vibration/rotation only through electronic spectra. [S007·5] ★★

No vibrational selection rule during electronic transitions; most start at v″ = 0; lines labelled (v′,v″) form a progression converging at high energy (anharmonic excited state). [S007·6-8] ★★★

Statement: *Franck–Condon principle: "An electronic transition takes place so rapidly that a vibrating molecule does not change its internuclear distance appreciably during the transition." Hence transitions are vertical.* [S007·13] ★★★

Case (a) $r_e' = r_e''$: (0,0) strongest. Case (b) $r_e' > r_e''$: strongest line (v′,0) with v′>0. Case (c) $r_e' \gg r_e''$: transition goes beyond the dissociation limit → continuum. [S007·15-16] ★★★

Why: A continuum means dissociation: the atoms fly apart with unquantised kinetic energy. The excited state usually has a longer bond length because of antibonding character. [S007·10,16] ★★★

Graphs

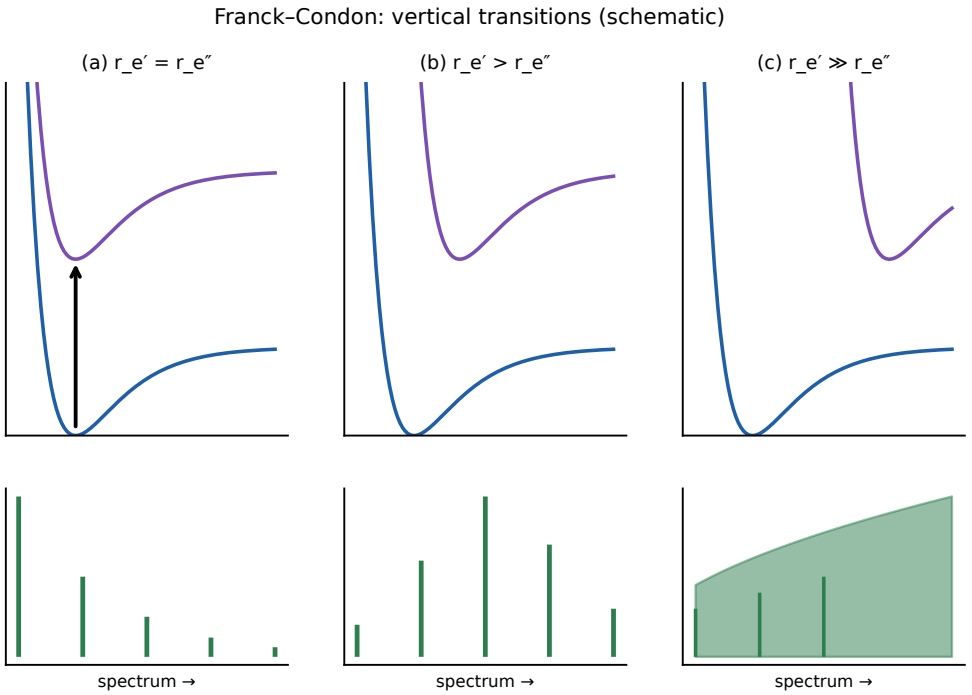


Fig. G21 — Franck–Condon, three cases (a, b, c) with spectra.

In an exam: Draw vertical arrow from v″=0; (c) ends in a continuum.

Flowcharts & tables

F7 · Franck–Condon outcome

	Compare r_e' with r_e''
↓	
	Equal → (0,0) strongest
↓	
	Longer → (v′,0), v′>0 strongest
↓	
	Much longer → continuum (dissociation)

Past-paper questions

2023–25 (×5) State the Franck–Condon principle; draw the three cases with spectra; meaning of a continuum.

M2 · Phase Equilibria

T8 · Basic terms & phase rule

Short note

Phase = homogeneous, separable part; component = min. independent constituents; F = min. intensive variables. F = C−P+2.

Formulas

F29 · Phase rule ★★★

$$F = C - P + 2$$

Gives Degrees of freedom · **Applies** no external fields · **Units** — · [S008·12]

Important points

Definition: Phase (P): homogeneous, physically distinct, mechanically separable part of a system bounded by a surface. Gases: P = 1; miscible liquids: P = 1; each solid / polymorph is a phase. [S008·4] ★★★

Definition: Component (C): smallest number of independently variable constituents by which the composition of every phase can be expressed (zero/negative values allowed). [S008·6] ★★★

Definition: Degrees of freedom (F): minimum number of intensive variables (T, P, concentration) to be fixed to define the system completely. [S008·10] ★★★

Statement: *Gibbs phase rule: provided equilibrium is not influenced by external forces (gravity, electrical, magnetic), $F = C - P + 2$.* [S008·12] ★★★

Counting: ice/water/vapour P=3, C=1; $CaCO_3 \rightleftharpoons CaO + CO_2$: P=3, C=2; $NH_4Cl \rightleftharpoons NH_3 + HCl$: P=2, C=1 (C=2 if extra HCl/NH_3 added); $Na_2SO_4 \cdot 10H_2O$ system C=2. [S008·5-9] ★★

Systems minimise Gibbs energy at constant T, P; at equilibrium G is equal in coexisting phases. [S008·3] ★

Flowcharts & tables

F9 · Phase-rule application

Count P
↓
Count C (minimum constituents)
↓
Gas absent & constant pressure? → F=C−P+1, else F=C−P+2
↓
Classify F=2 bivariant, 1 univariant, 0 invariant

Past-paper questions

(×2) Components and phases for $CaCO_3/MgCO_3$ decomposition.

T9 · Derivation of phase rule

Short note

Variables P(C−1)+2, constraints C(P−1), equal μ in all phases → F = C−P+2.

Formulas

F30 · Variables & constraints ★★★

$$\text{Variables} = P(C-1)+2 \; ; \; \text{constraints} = C(P-1) \; ; \; F = P(C-1)+2 - C(P-1) = C-P+2$$

Gives Derivation · **Applies** — · **Units** — · [S009·7]

F31 · Equilibrium condition ★★★

$$\mu_i^{\alpha} = \mu_i^{\beta} = \mu_i^{\gamma} = \dots$$

Gives Equal chemical potential · **Applies** each component · **Units** J mol^{-1} · [S009·3,7]

F32 · Chemical potential Transcript only ★★

$$\mu_i = (\partial G/\partial n_i)_{T,P,n_j}$$

Gives Partial molar Gibbs energy · **Applies** — · **Units** J mol^{-1} · [T008 (S009·5 unreadable)]

Important points

At equilibrium all phases have the same T and P and the chemical potential of each component is equal in every phase; mass flows from higher to lower μ. [S009·3-4] ★★★

Derivation: variables = P(C−1) composition terms + 2 (T,P); constraints = C(P−1); F = C−P+2. [S009·7] ★★★

Flowcharts & tables

F8 · Phase-rule derivation

Variables = P(C−1)+2

↓

Constraints = C(P−1)

↓

F = variables − constraints = C−P+2

Past-paper questions

2023–25 (×3) Derive the phase rule for C components and P phases.

T10 · One-component system: water

Short note

F = 3−P. Areas ice/water/vapour F=2; lines OA, OB, OC F=1; triple point O F=0 (0.0098 °C, 4.58 mmHg); critical 374 °C, 218 atm; OC slope negative; OA′ metastable.

Formulas

F33 · One-component ★★★

$$F = 3 - P$$

Gives Area F=2, line F=1, point F=0 · **Applies** — · **Units** — · [S009·8]

F34 · Clausius–Clapeyron ★★★

$$dP/dT = \Delta H/(T\Delta V)$$

Gives Slope of curve · **Applies** fusion, vaporisation, sublimation · **Units** Pa K^{-1} · [S009·12-14]

Important points

Water: F = 3−P → one phase F=2 (bivariant area), two phases F=1 (univariant line), three phases F=0 (invariant point). [S009·8-10] ★★★

Triple point O: 0.0098 °C, 4.58 mmHg (F=0). Critical point A: 374 °C, 218 atm; above it water cannot be liquid. [S009·15] ★★★

Why: Fusion curve OC has negative slope: ice melts with decrease in volume (ΔV<0 → dP/dT<0), and it is steep because |ΔV| is small. Sublimation (OB) and vaporisation (OA) slopes are positive and less steep because ΔV is large. [S009·12-14; T009b] Transcript-assisted ★★★

Metastable OA′: supercooled water; its vapour pressure is higher than that of ice at the same T; disturbance returns the system to stable OB. [S009·16] ★★★

Graphs

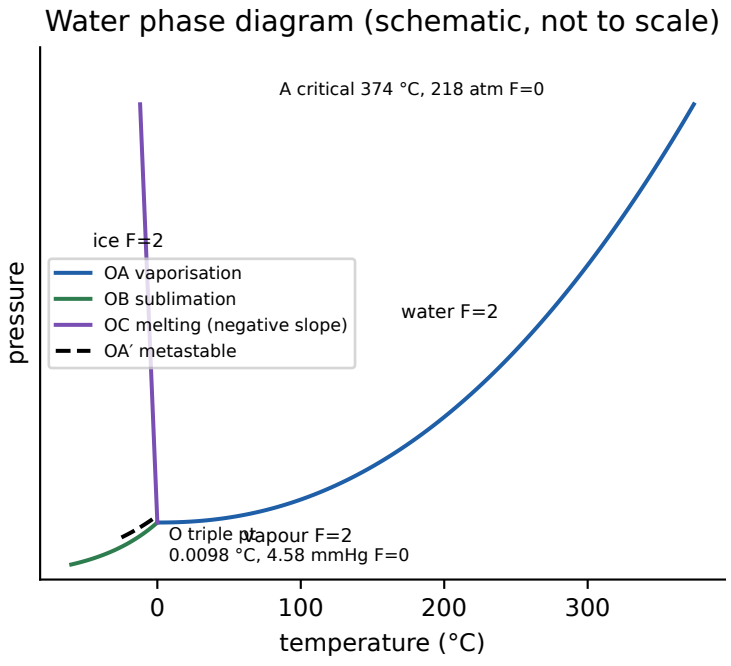


Fig. G22 — Water phase diagram (schematic, not to scale).

In an exam: Label OA, OB, OC, OA′, O, A, with T,P values and F in each region.

Past-paper questions

2023–25 (×4) Draw and label the water phase diagram; F in each region; metastable curve; negative slope of melting curve.

T11 · Two-component: reduced phase rule & Pb–Ag

Short note

Condensed system at fixed P: F = C−P+1. Pb–Ag eutectic: 303 °C, 2.6 % Ag, F=0; curves F=1.

Formulas

F35 · Reduced (condensed) rule ★★★

$$F = C - P + 1$$

Gives Constant pressure · **Applies** condensed systems · **Units** — · [S010·2-3]

Important points

Why: General phase rule not used for 2 components: C=2 gives F=4−P, F=3 for P=1, needing 3-D; so pressure is fixed (≈1 atm) → F = C−P+1. [S010·2-3] ★★★

Definition: Condensed system: gas phase absent, hardly affected by small pressure changes. [S010·3] ★★

Pb–Ag: components miscible in all proportions in melt, immiscible as solids; plot is temperature vs composition at constant pressure. [S010·4] ★★★

Freezing-point curves: C=2, P=2 → F=1 (univariant). Eutectic point O: P=3, F=0 (invariant), lowest freezing point, definite composition and sharp melting point. [S010·7-8] ★★★

Numerical facts: Pb mp 327 °C, Ag mp 961 °C, eutectic 303 °C at 97.4 % Pb / 2.6 % Ag. [S010; PDF] ★★★

Desilverisation of argentiferous lead (Pattinson-type process) exploits this: Pb crystallises first, leaving Ag-rich melt. [T010] Transcript only ★

Graphs

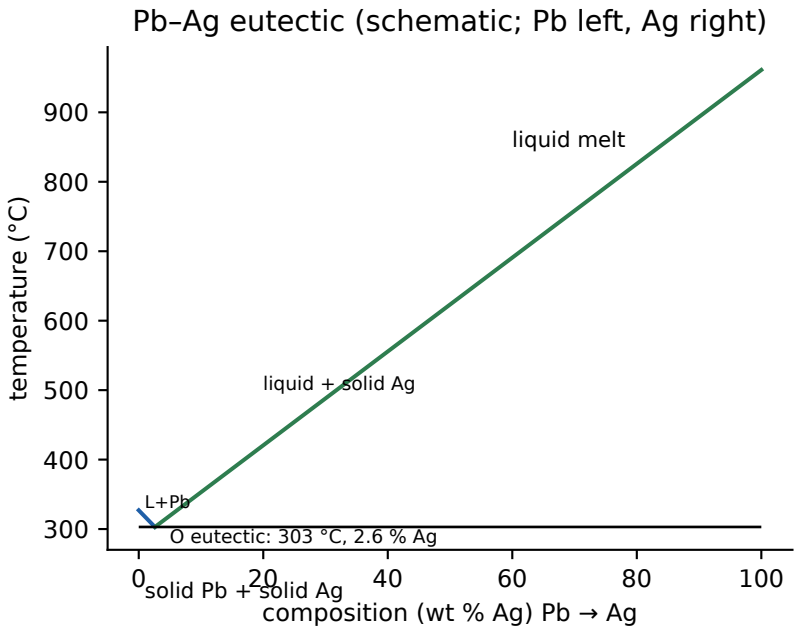


Fig. G23 — Pb–Ag eutectic diagram (schematic; Pb left, Ag right).

In an exam: Eutectic O at 303 °C, 2.6 % Ag; label the four areas. Slide lettering is inconsistent (see conflicts) – follow your faculty's.

Flowcharts & tables

F10 · Pb–Ag cooling path

	Melt cools → Pb separates, melt gets richer in Ag along curve to O
↓	
	At O (303 °C): Pb + Ag + melt, F=0
↓	
	Below O: solid Pb + solid Ag

Past-paper questions

2023–25 Draw the Pb–Ag diagram with eutectic values; why F = C−P+1.

T12 · Thermal analysis / cooling curves

Short note

Cooling curves: halt for pure/eutectic, slope break for mixtures; plot to build phase diagram.

Important points

Definition: Thermal analysis: study of cooling curves (temperature vs time) of various compositions during solidification; freezing and eutectic points are read from them. [S012·2] ★★

Pure substance: smooth fall, then a flat halt at the freezing point. Mixture: a change of slope when crystallisation starts, then a halt at the eutectic temperature. Plotting break/halt temperatures against composition builds the phase diagram. [S012·2-3] ★★

Graphs

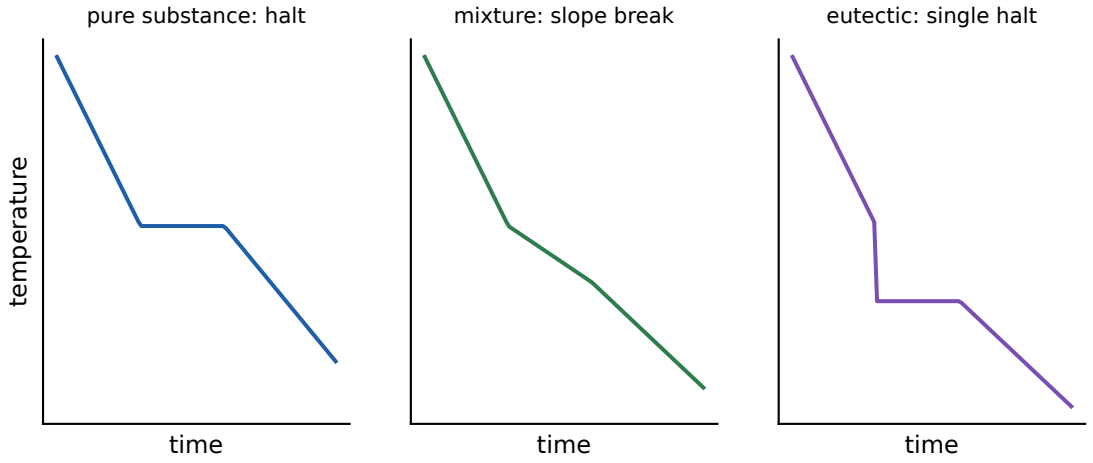


Fig. G24 — Cooling curves: pure, mixture, eutectic.

In an exam: Mark break and halt.

Flowcharts & tables

F11 · Thermal analysis → phase diagram

	Cool melts of several compositions
↓	
	Record T vs time
↓	
	Read break & halt temperatures
↓	
	Plot vs composition → diagram

M3 · Computational Chemistry

T13 · Computational chemistry

Short note

Computational chemistry simulates molecules: MM/MD (classical), semi-empirical, ab initio, DFT (quantum). Cheap, safe; AI adds speed.

Important points

Definition: Computational chemistry: techniques to solve chemical problems by simulating chemical systems, giving atomic-level information, using theoretical chemistry in efficient programs. [S011·3] ★

First theoretical work: Heitler and London, 1927. (Nobel 1998 Kohn & Pople; 2013 Karplus, Levitt & Warshel.) [S011·3; brief] Added for clarity ★

Scope: molecular properties (HOMO/LUMO, UV/IR/NMR spectra), reaction mechanism and kinetics, material design, drug discovery. [S011·6] ★

Classical: molecular mechanics (balls and springs, very fast) and molecular dynamics (motion of atoms). Quantum: semi-empirical, ab initio (wavefunction) and DFT (electron density; faster than ab initio, slower than semi-empirical). [S011·7-12] ★

Advantages: easier, cheaper, safer than experiments; complements, not replaces, experiment. [S011·13] ★

Ethylene: sp^2 C; σ + π bond; HOMO = π, LUMO = π*. Steps: geometry → basis set → method → solve → visualise. [S011·14-17] ★

AI-enhanced: fast screening, property prediction, reaction-pathway mapping, structure generation; speed-up from hours/days to milliseconds after training. [S011·19-20] ★

Graphs

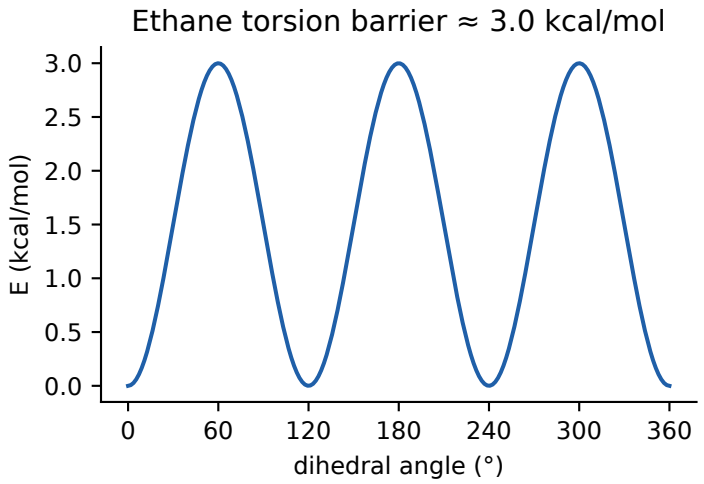


Fig. G25 — Ethane torsion barrier ≈3.0 kcal/mol (illustrative).

In an exam: Staggered minima, eclipsed maxima.

Flowcharts & tables

F12 · Computational methods tree

- Classical: molecular mechanics, molecular dynamics (fast, lower accuracy)
- ↓
- Quantum: semi-empirical → DFT → ab initio (slower, more accurate)

F13 · Ethylene electronic structure

- Define geometry
- ↓
- Select basis set
- ↓
- Apply method
- ↓
- Solve equations → wavefunction/density
- ↓
- Visualise HOMO, LUMO

F14 · AI-enhanced workflow

- Quantum simulations
- ↓
- Dataset
- ↓
- Train ML model
- ↓
- Predict properties in ms

MM / MD / semi-empirical / ab initio / DFT

Method	Basis	Speed	Accuracy
MM	classical springs	fastest	lowest
MD	laws of motion	fast	low
Semi-empirical	Schrödinger, approximated	fast	medium
DFT	electron density	medium	high
Ab initio	wavefunction	slowest	highest

Past-paper questions

(×1) Define computational chemistry; three applications.

Appendix A · Symbol glossary

Symbol	Meaning in this unit	Unit	Value / note
λ	wavelength	m, nm, Å	
ν	frequency	Hz	
$\bar{\nu}$	wavenumber	cm^{-1}	
c	speed of light	$m\ s^{-1}$ / $cm\ s^{-1}$	$3\times10^{8}\ m\ s^{-1} = 3\times10^{10}\ cm\ s^{-1}$
h	Planck constant	J s	6.626×10^{-34} (slides also use 6.625)
E, ε	energy; ε = energy in cm^{-1} (also molar absorptivity in Beer–Lambert — 3rd meaning!)	J; cm^{-1}; $L\ mol^{-1}\ cm^{-1}$	
ΔE	energy change	J	
J	rotational quantum number	—	0,1,2…
B	rotational constant = $h/8\pi^2Ic$	cm^{-1}	
I	moment of inertia	$kg\ m^2$	
μ	reduced mass $m_1m_2/(m_1+m_2)$ — ALSO chemical potential in T8–T9	kg; $J\ mol^{-1}$	1 amu = 1.66×10^{-27} kg
m_1, m_2	atomic masses	kg	
r_o	bond length (rigid rotor)	m, Å	1 Å = 10^{-10} m
r_1, r_2	distances of atoms from centre of mass	m	
ω	angular velocity	$rad\ s^{-1}$	
L	angular momentum	$kg\ m^2\ s^{-1}$	
k	force constant	$N\ m^{-1}$	
x	displacement	m	
v	vibrational quantum number	—	0,1,2…
v_osc, $\bar{\nu}$_osc	oscillation frequency / wavenumber	Hz ; cm^{-1}	
$\bar{\omega}_e$	equilibrium vibrational wavenumber (anharmonic)	cm^{-1}	
x_e	anharmonicity constant	—	small, positive
D_eq	depth of Morse well	J	
a	Morse constant	m^{-1}	
r_eq	equilibrium bond length	m	
r_e′, r_e″	bond length in excited / ground state	Å	
v′, v″	vibrational number in excited / ground state	—	
ε′, ε″	energy in excited / ground state	cm^{-1}	
F	degrees of freedom (variance)	—	
C	number of components	—	
P	number of phases — ALSO pressure (clash!)	— ; atm, mmHg	
T	temperature	K, °C	
G	Gibbs energy	J	

Symbol	Meaning in this unit	Unit	Value / note
n_i	moles of component i	mol	
ΔH	enthalpy of transition	J mol^{-1}	
ΔV	volume change of transition	m^3 mol^{-1}	
A	absorbance	—	
I_o, I	incident / transmitted intensity	—	
c, l	concentration, path length (Beer–Lambert)	mol L^{-1}, cm	
N_A	Avogadro constant	mol^{-1}	6.022×10^{23}

Appendix B · Constants

Constants: h = 6.626×10⁻³⁴ J s · c = 3×10⁸ m s⁻¹ = 3×10¹⁰ cm s⁻¹ · N_A = 6.022×10²³ mol⁻¹ · 1 amu = 1.66×10⁻²⁷ kg · π = 3.14 (papers). **Conversions:** E(J) ÷ hc → cm⁻¹; ν = c·ν̄; 1 eV = 1.602×10⁻¹⁹ J = 8065.5 cm⁻¹; 1 kJ mol⁻¹ = 83.59 cm⁻¹.

Appendix C · Possibly examinable, not in current material

Not on current slides **Fe–C diagram**: named in the 2025 module list, asked in 2024 ESAs (peritectic, eutectoid points); the class 010 transcript says it is the next class; no 2026 deck exists. **CO_2 phase diagram**: asked once (Jan 2024), not in current material.

Appendix D · Sources & gaps

Used: 12 slide decks (.pptx rendered, .slides.md, .slides.txt), 13 transcripts' text, 137-page Unit-1 PDF, course metadata. **Gaps:** class 003 video failed to download (HTTP 401) so no transcript; class 012 has no video. No videos were consulted. See README.md for what was not done.